Power Screws

Concepts

- A **power screw** (or lead screw) converts rotation into linear motion. They are often used to generate large axial output force to raise or lower a load via a rotational input torque. Examples include screw jacks and linear stages.
- In this class, we only consider power screws with **loads acting parallel** to the screw axis. We **lower** the load by rotating the screw to move the load in the same direction as the load, and **raise** the load by rotating the screw to move the load in the opposite direction as the load.
- For a square or acme power screw with major diameter d, pitch p, and n starts, then its **minor (or root) diameter** d_r, **pitch (or mean) diameter** d_m, **lead** ℓ, **thread width** w, and **thread depth** t are given by:

$$d_r = d - p \qquad d_m = d - p/2 \qquad \ell = np \qquad w = p/2 \qquad t = p/2$$

- The **thread angles** for square and acme threads are $\alpha_{sq} = 0°$ and $\alpha_{ac} = 14.5°$ respectively.
- For a power screw with pitch diameter d_m, thread angle α, lead ℓ, coefficient of friction f, and applied load F, the **raising torque** T_R and **lowering torque** T_L are respectively:

$$T_R = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right) \qquad T_L = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha - \ell}{\pi d_m + \ell f \sec\alpha}\right)$$

- A power screw is **self-locking** when T_L is positive, i.e., when $\ell < \pi d_m f$.
- If a power screw with applied load F has a collar with collar diameter d_c and coefficient of friction f_c, the **collar torque** T_C is given by:

$$T_C = F f_c d_c / 2$$

- The **efficiency to raise a load** via a power screw is the ratio of the total torque in the absence of friction to the total torque in the presence of friction, and is given by:

$$e_R = \frac{F\ell}{2\pi(T_R + T_C)}$$

Practice Problems

1) A triple-threaded Acme power screw has a major diameter of 20 mm and a minor diameter of 15 mm. How many meters will the power screw move axially if it is rotated by two full revolutions?

2) It is observed that a given square-thread power screw with 6 cm diameter and 1 cm pitch requires a 50 N·m minimum applied torque in order to lower a 10 kN load. Estimate the coefficient of friction between the power screw and its nut. Is this power screw self-locking?

3) You are tasked with designing a single-threaded Acme power screw with collar diameter and screw pitch that are double and one fifth the major screw diameter respectively. Assuming the coefficients of friction for the collar and screw are equal, compute a coefficient of friction that will achieve a 30% raising efficiency.